GENAI BASICS • LESSON 07 • THE COPILOT ECOSYSTEM

One Copilot? No: a family of three

How a team ships one feature in a week, with Copilot helping at every step

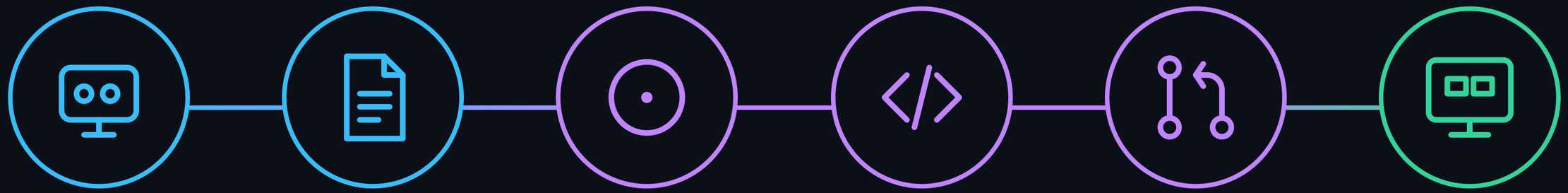
3 Copilots • 1 team • 1 feature • 1 week. All data is synthetic. Free tiers are enough for practice.



THE STORY

One feature goes through six steps in a week

Every step produces **a text you check yourself**.



Teams	Word	GitHub issue	VS Code	Pull request	Windows
“Prime: free delivery from $75, city only”	One-page brief: decisions and open questions	#12 Free delivery for Prime from $75	+2 lines of logic, +3 tests, all 7 green	Review: “74.99 test is there; use an exact type for money”	Checklist item 6: roll back the flag if outside-zone fees change

THE MAP

Three Copilots, three places where your information lives

To pick a Copilot, answer one question: **where does the information live?**



Sees: code, history, issues and pull requests



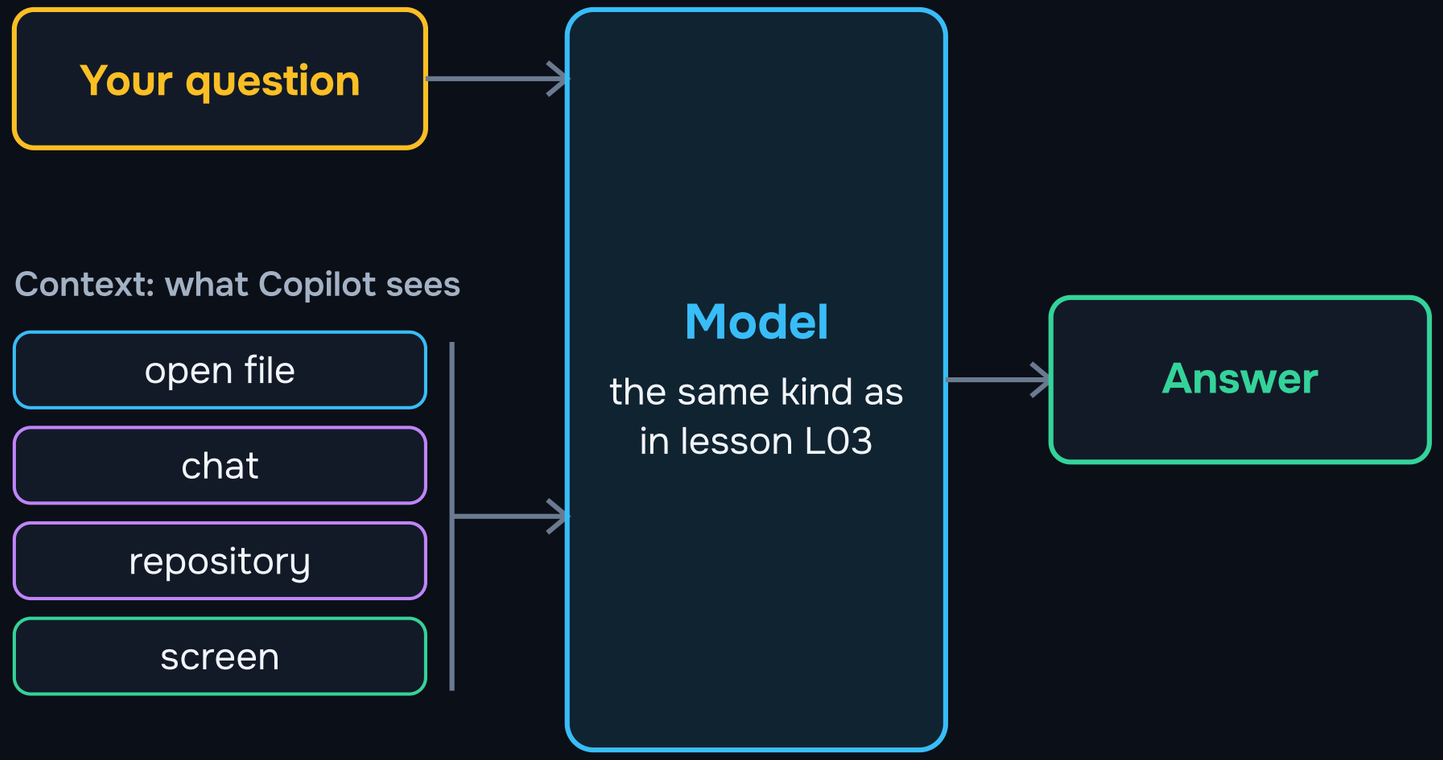
Sees: mail, files and meetings that are open to you



Sees: the window you shared, and your files

HOW IT WORKS

Same model as in L03: what matters is what it sees



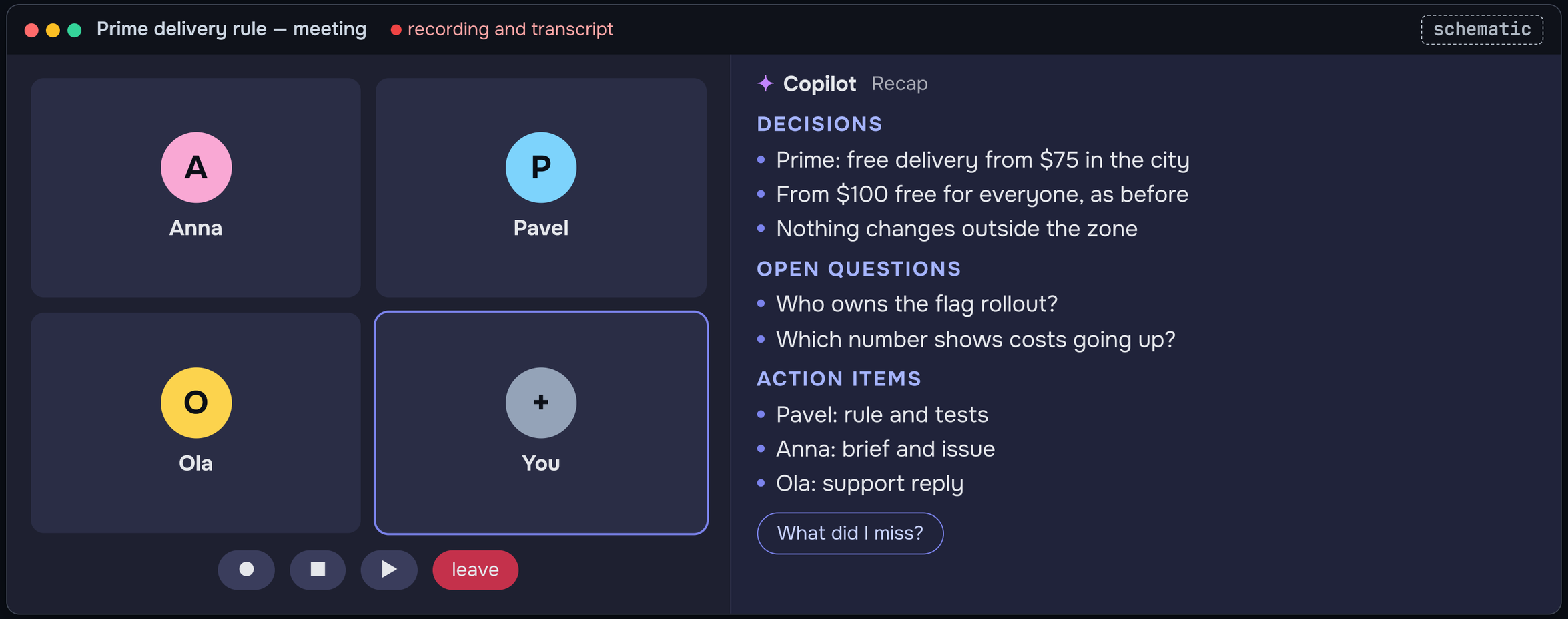
YOU ALREADY DID THIS IN COLAB

```
# L03
response = client.models.generate_content(
    model=MODEL,
    contents=question + file_text)
```

> Copilot makes this call itself and **adds your file** to the request. More of the right context, a better answer.

TEAMS

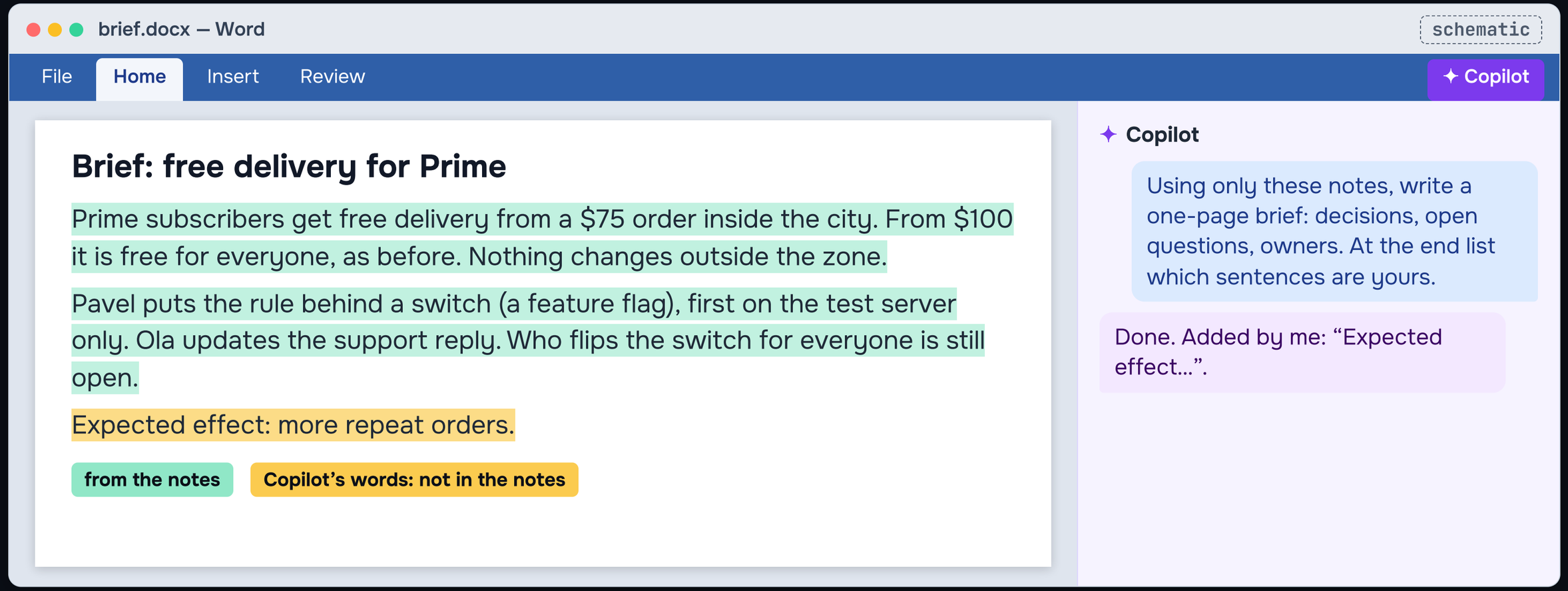
Teams: a meeting becomes decisions and action items



You need a meeting transcript and licensed Copilot. No access: paste `meeting_notes.md` into free Copilot Chat.

WORD

Word: from notes to a one-page brief



Outlook and Excel: the same button, a different context

Outlook — support reply

schematic

Ola
Customers ask why some Prime orders still pay for delivery.

Anna
Because the rule is city only, from $75.

Ola
Can we put that in one sentence in the support reply?

✦ Summary
Prime: free from $75, city only. Ola wants one sentence for the support reply.

orders.xlsx — Excel

schematic

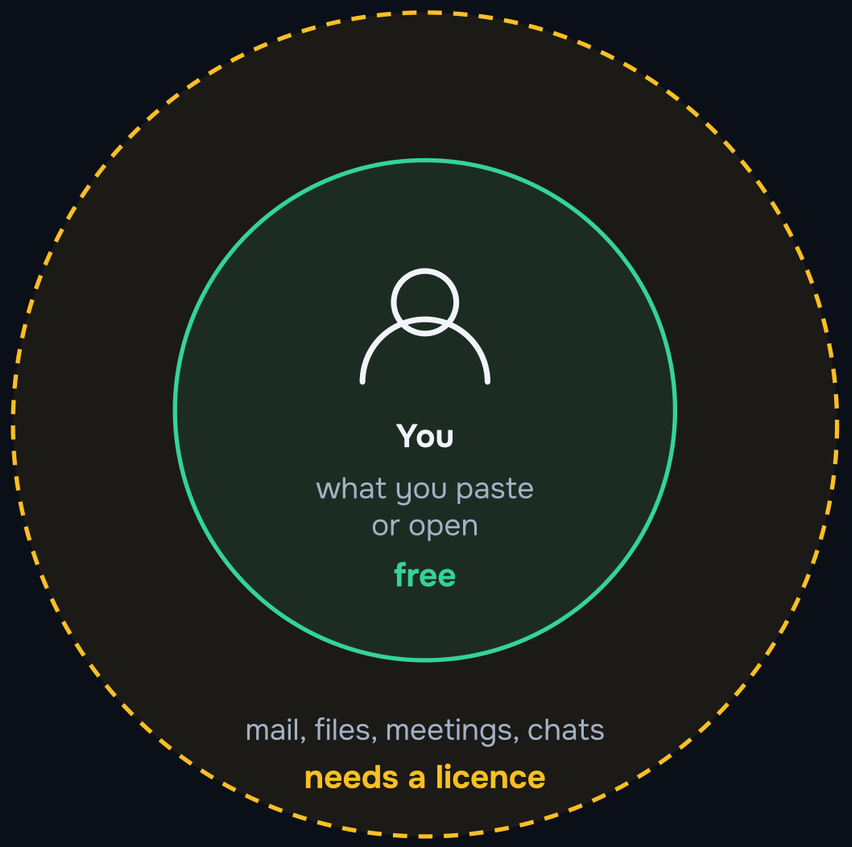
order_id	zone	is_prime	subtotal
1001	CITY	TRUE	82.40
1002	CITY	FALSE	76.00
1003	OUTSIDE	TRUE	91.10
1004	CITY	TRUE	75.00
1005	OUTSIDE	FALSE	48.30
1006	CITY	TRUE	120.00
1007	CITY	TRUE	64.90
1008	OUTSIDE	TRUE	79.50

✦ Copilot
Most Prime orders from $75 are in zone CITY: 3 of 5.

MONEY **OCTOBER 2026 · US PRICES, USD**

What is free and what is paid

Free: the web and what you give it yourself. Licensed: your mail, files and meetings within your permissions.



GitHub Copilot

Free	limited completions and chat per month	**$0**
Pro	per month	**$10**
Pro+	per month	**$39**
Business	per user per month	**$19**
Enterprise	per user per month	**$39**

Microsoft

Copilot Chat	with a Microsoft 365 work account: the web and what you open or attach	**included**
Business	Microsoft 365 Copilot, billed yearly	**$23.50**
Enterprise	same, billed yearly	**$30**

GITHUB

GitHub issue: the brief becomes a task for the developer

courier-app / Issues

schematic

Free delivery for Prime customers from $75 #12

Open Anna opened this issue

- Prime with an order **of $75 or more**: free delivery **in the city**
- Free **from $100** for everyone, as before
- Outside the zone **nothing changes** (fee $18.00)
- Boundary tests: 75 and 74.99, and one non-Prime test
- The error for a negative subtotal stays

Acceptance: all tests pass, the change is small.

Assignees

P Pavel

Labels

feature

Copilot

Assign to Copilot

Cloud agent: not on Free. Covered in L16.

THE EDITOR

In the editor Copilot helps in three ways

1 Completion finishes your line, **2 chat** answers about the open file, **3 agent** edits files and runs tests itself.



THE RESULT

The result: a 2-line diff and 7 green tests

This is a real run, not a drawing. A diff is the exact list of added and removed lines. Look at it and at the tests, not at the word “done”.

GIT DIFF -U1

```
delivery_fee.py
@@ -16,2 +16,4 @@ def delivery_fee(order: Order) -> float:
         return 0.0
+    if order.zone.upper() == "CITY" and order.is_prime and order.subtotal >= 75:
+        return 0.0
     if order.zone.upper() == "CITY":
test_delivery_fee.py
@@ -14,2 +14,11 @@ class DeliveryFeeTests(unittest.TestCase):

+    def test_prime_free_at_75(self):
+        self.assertEqual(delivery_fee(Order(75, "CITY", True)), 0.0)
+
+    def test_prime_below_75_pays(self):
+        self.assertEqual(delivery_fee(Order(74.99, "CITY", True)), 7.50)
+
+    def test_non_prime_at_75_pays(self):
+        self.assertEqual(delivery_fee(Order(75, "CITY", False)), 7.50)
+
     def test_negative_subtotal_rejected(self):
```

PYTHON -M UNITTEST -V

```
test_city_standard ... ok
test_free_delivery_from_100 ... ok
test_negative_subtotal_rejected ... ok
test_non_prime_at_75_pays ... ok
test_outside_standard ... ok
test_prime_below_75_pays ... ok
test_prime_free_at_75 ... ok

Ran 7 tests
OK
```

7 tests ok

+2 lines of logic

+3 new tests

Test names shortened, the rest untouched. Boundaries: 75, 74.99, non-Prime.

REVIEW

Copilot review: a second pair of eyes, not the judge

courier-app / Pull requests

schematic

Free delivery for Prime from $75 #13

Conversation Commits 1 Files changed 2

delivery_fee.py +2 -0

```
@@ -16,2 +16,4 @@ def delivery_fee(order: Order) -> float:
         return 0.0
+    if order.zone.upper() == "CITY" and order.is_prime and order.subtotal >= 75:
+        return 0.0
     if order.zone.upper() == "CITY":
```

Copilot review

The 74.99 boundary test is already there. Consider `Decimal`, an exact type for money, instead of float.

Pavel reply

Decimal goes into a separate issue #14; here I keep float, like the rest of the file.

test_delivery_fee.py +9 -0

Approved: Ola
a person decides

Checks: tests 7/7
tests are the gate

Copilot: 1 comment
a suggestion, not an order

TEAM RULES

copilot-instructions.md: a README for the robot

A short file of team rules. Copilot reads it on every chat request and review.

.github/copilot-instructions.md	where it shows in the result
1 **Keep the code small and explicit.**	→ 2 lines of logic in the diff
2 **Use only the Python standard library; add no dependencies.**	→ new imports: 0
3 **Keep current behavior unless the issue says to change it.**	→ `test_outside_standard`: fee still $18.00
4 **Add boundary tests for every price threshold.**	→ `test_prime_below_75_pays`: 74.99 pays $7.50
5 **Explain any assumption before you edit code.**	→ the plan in chat before the edit (slide 10)

GITHUB COPILOT MAP

Four doors to one GitHub Copilot; today we used two

Editor

VS Code, JetBrains, Visual Studio

Pavel: “Do ISSUE.md and run the tests”

TODAY

on Free

github.com

issues and pull requests

Anna: “Copilot, review #13”

TODAY

depends on the plan

Terminal

Copilot CLI

“Show me which tests failed”

L16

on Free

Cloud

cloud agent

Assign issue #12 to Copilot → it opens a PR

L16

paid plans

BREAK

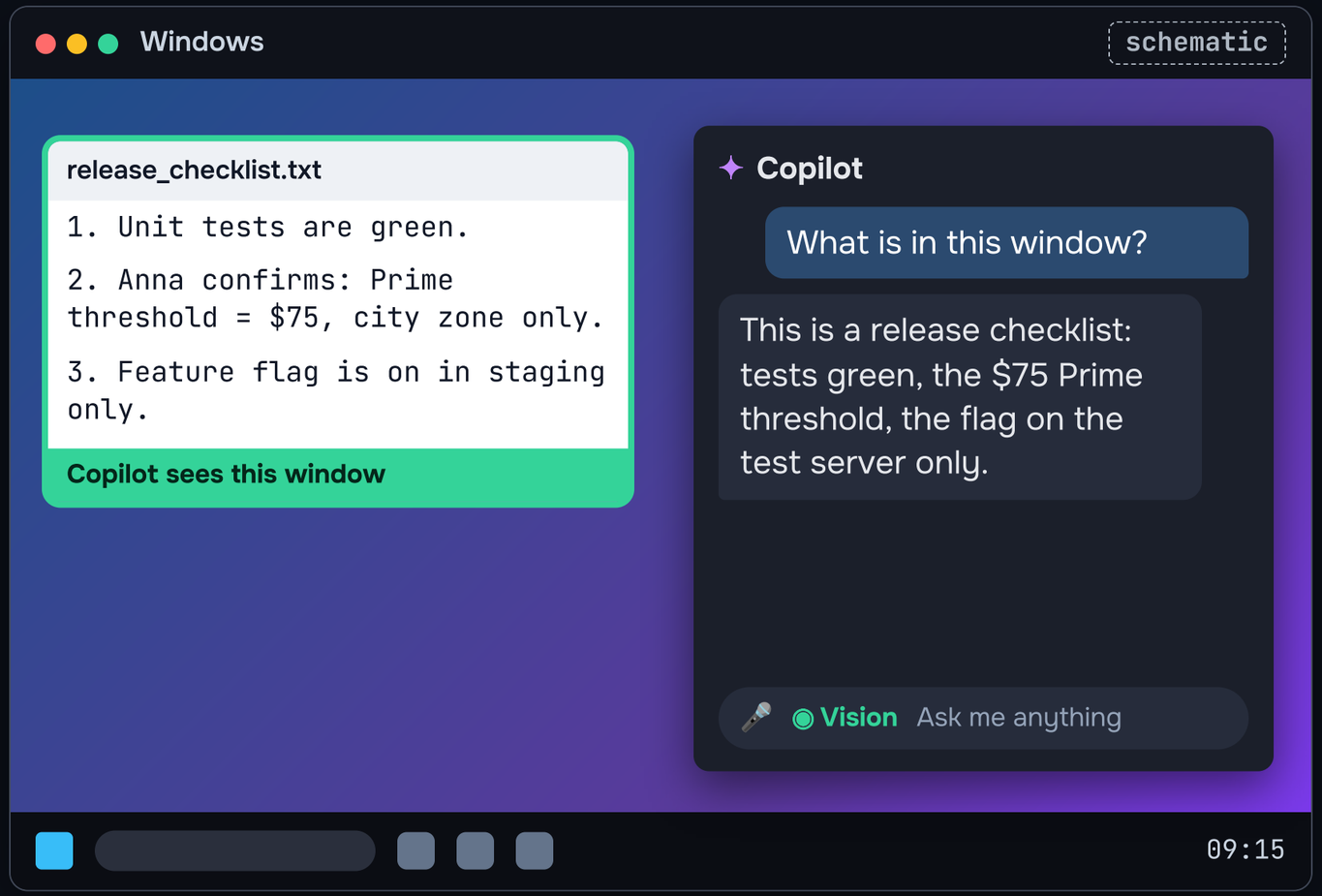
5 minutes: coffee and a rest from the screen

05:00

Next: the Copilot key in Windows, the release checklist, six checks and homework.

WINDOWS

Windows: Copilot sees only the window you shared





- Chat and voice
- Vision: sees the window you pick
- File search: docx, xlsx, pptx, txt, pdf
- Screenshot of the screen, then a question about it

Work accounts open the web chat instead of the app.

RELEASE DAY

Release day: Copilot reads the open checklist

release_checklist.txt — Notepad

schematic

Courier app: release checklist

1. Unit tests are green.
2. Anna confirms: Prime threshold = $75, city zone only.
3. Feature flag is on in staging only.
4. Ola updates the support reply before the production release.
5. Compare the delivery-cost number for 24 hours after the release.
6. Roll back the flag if outside-zone fees change unexpectedly.

Which item catches an outside-zone fee change?

✦ Item 6: “Roll back the flag if outside-zone fees change unexpectedly.”

> The file is open, so the answer is about the file. No Vision: **paste the text into the chat**.

ACCEPTANCE

Six checks are the same for any Copilot

This is also the homework rubric. On the right are the results of our week.

✓ The brief marks what is from the notes and what is invented	the effect line is marked
✓ The issue has numbers and what must not change	$75 · $100 · $18.00
✓ The change is small	+2 lines of logic
✓ Tests are green and check the boundary	7 of 7 · 75 and 74.99
✓ The review was read and answered	1 Copilot comment · Pavel’s reply
✓ Evidence is kept	slides 11, 12, 17 + checklist.md

LIMITS

Copilot sees only what you can open



Copilot is you, but faster

Wrong account
You signed in to the wrong one, and Copilot sees the wrong files.

A secret in the chat
A password, a key, customer data: none of it goes into a request.

“Done!”
The word proves nothing. Look at the diff and the tests.

Model training: work accounts (Microsoft 365, GitHub Business and Enterprise) are not used. Personal GitHub: since 24 April 2026 your data may be used by default; switch it off in Copilot settings.

PRACTICE

Homework: take your feature through three Copilots

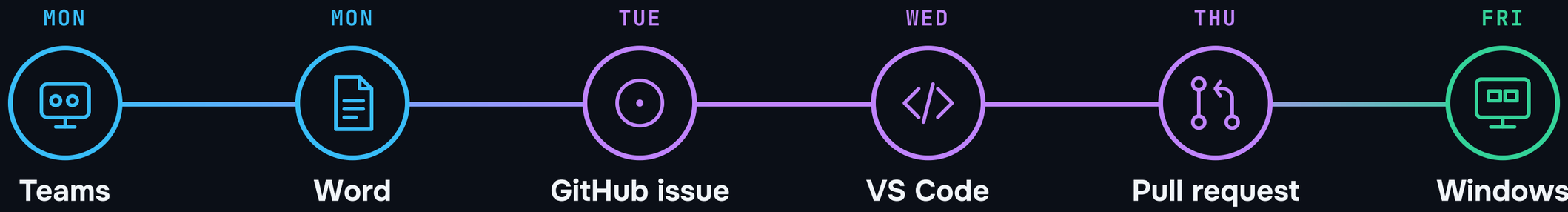
WHAT TO HAND IN: A PUBLIC GITHUB REPOSITORY

```
courier-copilot-week/
├── brief.md
├── delivery_fee.py
├── test_delivery_fee.py
├── checklist.md
└── README.md   # 3 screenshots
screenshots: brief, Copilot in editor, PR
+ a link to one merged pull request
```

- **Feature:** the Prime rule or your own, for example “rain: +$2 outside the zone”.
- **Tools:** free ones only: GitHub Copilot Free in VS Code and Copilot Chat.
- **First** switch off training on your data in GitHub settings.
- **Review:** Copilot review on a PR may be paid; a human comment that you answered is enough.
- **Grading:** the six checks of slide 18. Synthetic data only.

SUMMARY

Summary: three places, one habit: check the result



1. First ask: where does the information live?
2. Copilot sees only what is open to you.
3. Look at the diff and the tests, not at “done”.
4. Keep evidence: screenshots, a checklist.
5. No secrets or customer data in a request.

NEXT: LESSON 08

AI for data analysis

order_id	zone	is_prime	subtotal
1001	CITY	TRUE	82.40
1002	CITY	FALSE	76.00

Copilot opens the table from slide 7 and we do real analysis.